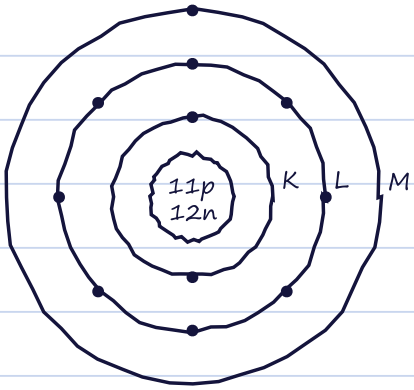
Name: Priya Nair Roll No: 1CH24CS027 Sec: B

Chemistry - Internal Assessment 1

Ans 1)

Sodium has 11 protons and 12 neutrons in the nucleus.

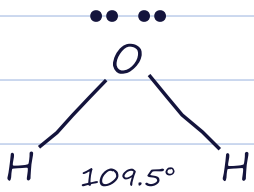


Sodium (Na)

Configuration 2, 8, 1. Valency is 1.

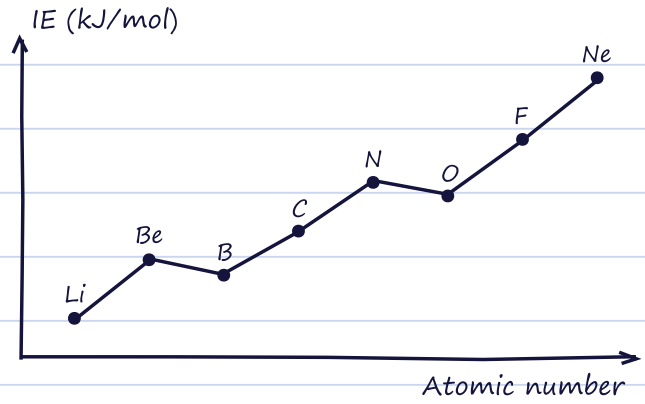
Ans 2)

Water has two bond pairs and two lone pairs on oxygen.



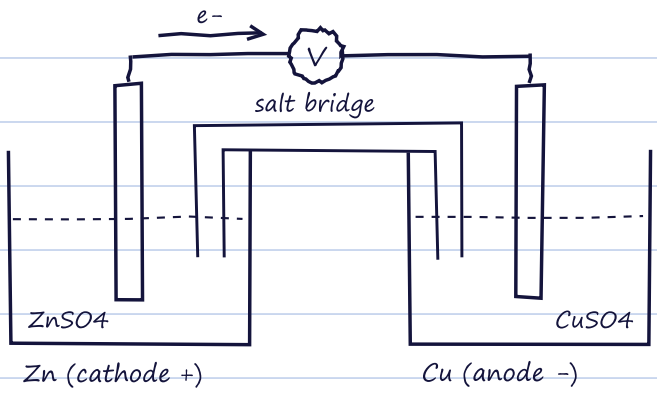
Bond angle is 109.5° because there are four electron pairs around oxygen. Lone pairs repel more strongly so the shape is bent.

Ans 3)



Ionisation energy increases across the period because the nuclear charge increases and the size decreases. B is lower because its electron is in 2p which is further from the nucleus. O is lower because of repulsion between paired 2p electrons.

Ans 4)

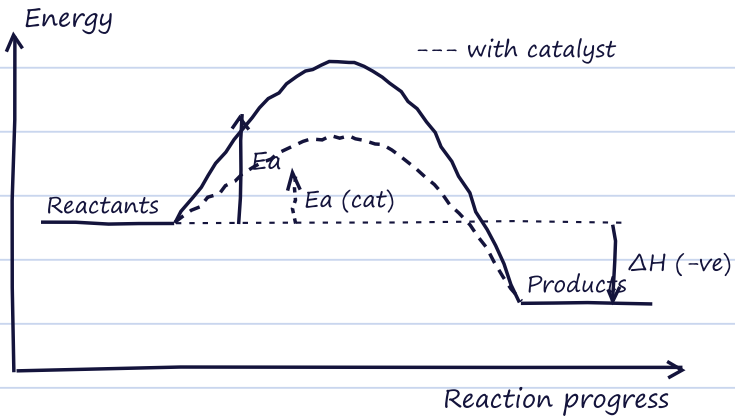


Anode: Cu -> Cu2+ + 2e-

Cathode: Zn2+ + 2e- -> Zn

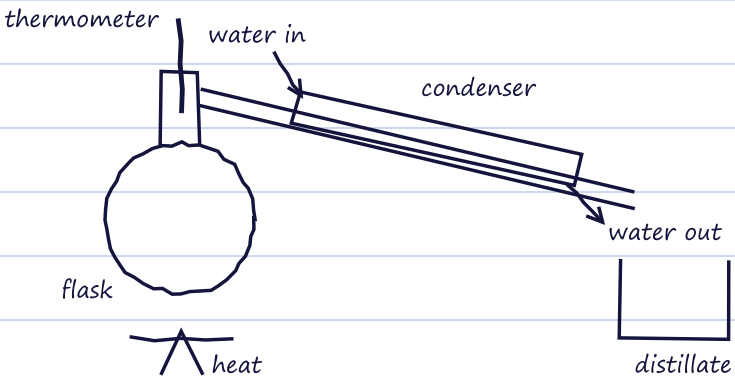
EMF = 0.34 + 0.76 = 1.10 V

Ans 5)



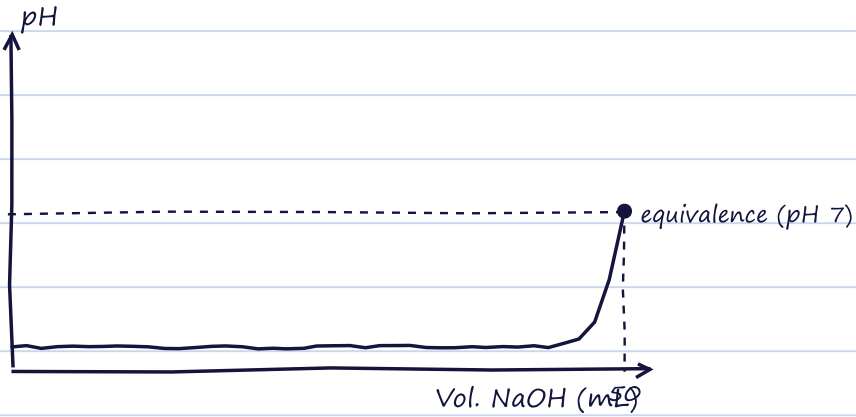
The catalyst lowers the activation energy by giving another pathway so more particles can react and the rate increases. dH stays the same.

Ans 6)



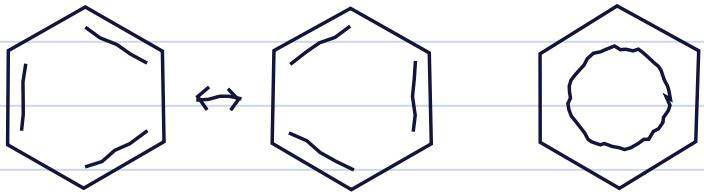
The condenser cools the vapour into liquid. Water goes in from the top so that it flows down with gravity and cools faster.

Ans 7)



Volume of NaOH = 0.10 x 25 / 0.05 = 50 mL. Equivalence point at pH 7. Indicator phenolphthalein.

Ans 8)

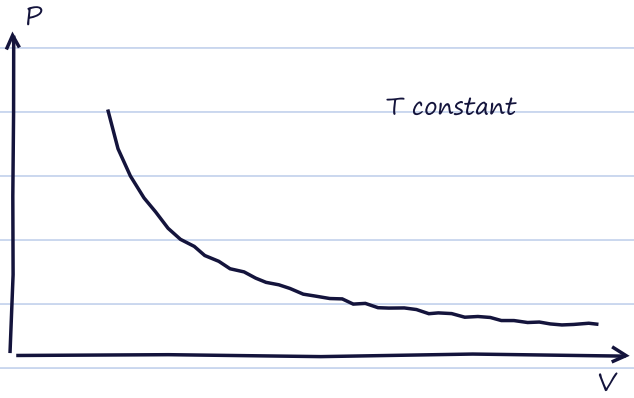


Kekule structures hybrid

Benzene has delocalised electrons that make it very stable, so it does substitution and keeps the ring stable instead of addition.

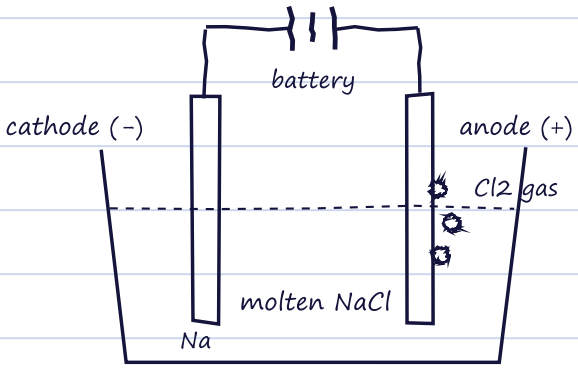
Ans 9)

Boyle's law: P is inversely proportional to V at constant temperature.



P1V1 = P2V2, so V2 = 6 x 3 / 2 = 9 L

Ans 10)



At cathode $Na^+ + e^- \rightarrow Na$

At anode $2Cl^- \rightarrow Cl_2 + 2e^-$

Sodium at cathode and chlorine at anode.